ALGORITHM

Smart Traffic Signal Scheduling System Using Stack and Queue

Algorithm: Traffic Signal Scheduling

1. Start.

2. Create an empty queue Q for traffic signals and an empty stack H for signal history.

3. Read the number n of signals. If n is not positive, display an error and stop.

4. For each signal, read its name and green duration. Reject an empty name or a non-positive duration.

5. Insert each valid signal at the rear of queue Q.

6. Display the queue from front to rear using a temporary copy.

7. Ask the user whether to start scheduling.

8. If scheduling is started, take the front signal from Q and remove it from the queue.

9. Display the selected signal as GREEN and show its duration.

10. Wait for the user to press Enter to complete the simulated interval, or enter q to stop.

11. If q is entered, restore the selected signal to the front of Q and stop scheduling.

12. Otherwise, display that the signal is now RED, push it onto history stack H, and insert it at the rear of Q.

13. Repeat Steps 8–12 until the user requests stop.

14. If requested, display history by reading a temporary copy of H from top to bottom.

15. End.

Data Structure Operations

Queue: front() reads the next signal, pop() removes it, and push() inserts it at the rear. **Stack:** push() records a completed signal and top()/pop() on a temporary copy displays the newest history entry first.

Example sequence

Input queue: North → East → South. After North completes its turn, history stack top is North and the queue becomes East → South → North. The same process repeats in FIFO order.